PAPERGRAPH / CRISPR OBSERVABILITY

Follow what is observed. Keep every inference conditional.

Question → contract → evidence → conditional conclusion → frontier.

Read stations 01–12. Dotted arrows mean “read next”, not “proves”.

01 QUESTION

What can the readout justify?

An editing readout needs an observation contract before it can support a product claim.

Methods: units; Table S1

02 DEFINITION

Declare event and denominator

Local peptide equivalence, exact programmed DNA and software MODIFIED are different events.

Methods: events; Tables 4A–4B

03 MECHANISM

The product can hide its anchor

The documented anchor no longer fits after displacement; the same-input contrast depends on the observation rule.

Table 1; Figure 1; Prop. 1

04 FORMAL DERIVATION

Retain unknowns; derive bounds

Retaining unknown mass yields sharp conditional extrema without assigning unobserved outcomes.

Table 2; Propositions 2–3

05 COUNTEREXAMPLE

Equal labels, different strings

Reference-sensitive labels and actual equal-label/unequal-string records refute simple interchangeability.

Results: correspondence; Figure 4

06 METHOD

Require a common observed string

The witness accepts only identical observed strings from three declared extractions.

Tables 2, S7; Proposition 4

07 COMPUTATIONAL RESULT

Read the conditional result

All three native witness PE-minus-mock lower endpoints are positive under the stated premises.

Table S8; Proposition 5

08 SENSITIVITY

Stable sign, changing labels

Finite-grid positivity and record-level stability are different properties.

Table S12; finite-grid Results

09 FORMAL DERIVATION

Preserve a declared target

Observed-catalog factorization is target-relative and requires constancy within representation fibers.

Supplement: joint targets / criterion

10 INTERPRETIVE CASES

Keep the counterexamples

Ambiguous targets and constant target support constrain what a successful certificate means.

Table 3; ambiguity / constant targets

11 REPRODUCIBILITY

Separate replay and independence

Source identity and computational reproduction support consistency within the declared evidence scope.

Methods: verification; replay levels

12 CONSEQUENCE AND FRONTIER

Report the result and frontier

The practical proposal is a claim-specific reporting contract with explicit unresolved frontiers.

Methods: contract; revision tests

SCIENTIFIC RELATIONS LIVE IN THE ARGUMENT MODEL

Requires jointly: stated assumptions. Derived under: the declared finite-record contract.

Supported by: an identified source and operation. Limited by: an unmeasured quantity or changed domain.

The overview is a reading map; open a station to inspect these typed links and the evidence they reference.

Conditional identification hull: [E/N, (E+U)/N] · Accepted labels must be correct.

No molecular-accuracy, prospective-validation or haplotype-function claim is supplied by this diagram.

Public diagram · formal text withheld · Facundo Firmenich · source-bound model 294d2259f24142ac